

Design and testing of a Hardware-in-the-Loop system for a grid integrated Ocean Current Turbine[☆]

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ABSTRACT

Renewable energy is an increasingly vital field that continues to evolve with growing demands for innovative electricity production methods. Among these, Ocean Current Turbines (OCTs) have emerged as a promising technology that targets the vast energy potential of ocean currents. Building on established Hardware-in-the-Loop (HIL) methodologies primarily used in wind turbine testing, this research adapts and extends these techniques to develop a HIL testbed for OCTs. This research involves the development of a robust simulation model, designed to replicate the dynamic ocean current conditions impacting an OCT. The simulation is built with Simulink and is executed within Opal-RT's real-time simulation environment. The simulation model consists of an OCT rotor model outputting desired shaft torque values to a dynamometer which uses measured torque and speed feedback to integrate with a micro-grid setup. This model aims to operate in both Region 2 and Region 3 of power generation, as well as the transitional Region 2.5. A variable-speed controller is developed to maximize the power capture in Region 2 and is evaluated on the HIL testbed, a blade-pitch controller is employed to prevent over-generation in Region 3 and is simulated inside the real-time environment, and both of these controllers are evaluated simultaneously while operating in region 2.5 on the HIL testbed. This study demonstrates a simple yet effective approach to HIL simulation for marine energy systems. The findings reinforce the feasibility of HIL testing for OCTs and contribute valuable insights into the broader context of marine renewable energy technology development.

1. Introduction

Renewable energy, and particularly marine hydrokinetic (MHK) energy, has become a prominent topic in the global quest for sustainable, low-carbon energy solutions. MHK energy harnesses the kinetic power of ocean currents, waves, and tides, providing a reliable and predictable renewable energy source [1]. By harnessing the potential of powerful ocean currents, ocean current turbines (OCTs) generate electricity as their blades rotate with the water flow, converting the mechanical motion of water into electrical energy. OCTs tend to be anchored to the ocean floor and must be designed to endure strong currents, corrosive saltwater, and minimal maintenance. Due to the consistency and

predictability of high-current streams such as the Florida Gulf Stream, OCTs offer a stable and promising solution for large-scale renewable energy production largely unaffected by most weather conditions [2].

OCT systems offer a variety of designs, with horizontal axial flow turbines standing out for their popularity due to their practicality and efficiency [3]. These turbines closely resemble traditional wind turbines and excel at capturing energy from ocean currents, making them a focal point for research and development in the field. Other designs, such as the Vertical Axis Hydro-turbine [4], also exist and have proven to be effective. Major ocean currents, such as the Florida Gulf Stream and the Kuroshio Current, possess significant energy potential. Studies by [5,6] evaluated the Gulf Stream's theoretically extractable power

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under various realistic scenarios. They found that the peak extractable power could range from 5.1 GW to as high as 40.9 GW, depending on factors such as turbine array area and estimated drag coefficients, highlighting the immense potential of these technologies as a renewable energy source.

The concept of harnessing ocean current energy gained momentum after the 1973 oil crisis, leading to studies like the 1974 McArthur workshop, which assessed the feasibility of marine hydrokinetic (MHK) energy in the Florida Gulf Stream and proposed prototype installations [3,7]. In 1980, William Mouton patented an early ocean current turbine (OCT) design for an underwater power plant, though material and engineering limitations hindered economic viability [8]. A 1985 study of a 4 kW Darrieus OCT in the Gulf Stream identified challenges such as high mooring costs, non-steady torque at low RPMs, and corrosion issues, resulting in estimated costs of \$2,000 per kW for a larger turbine [4]. Advances in material science and marine engineering later improved the feasibility of OCTs, building on similarities from marine current turbine (MCT) projects, such as the UK's 2003 installation of "Seaflo", a 300 kW tidal turbine, demonstrating the potential for practical OCT systems [9].

Unlike tidal turbines, OCTs primarily focus on being deployed in stable high-energy currents such as the Florida Gulf Stream or the Kuroshio Current, which makes them a candidate for base power with grid integration. However, these harsh environments make it difficult to prototype new OCT models in an economically viable way, thus a desire for accurate mathematical models and simulated environments becomes highly valued. A variety of mathematical models for OCTs have already been developed as shown in both [10,11], but purely simulated numerical models can cause errors and complexities such as power take-off (PTO) errors, transient response errors, and complex modeling of the structural dynamics of the OCTs body and actuators. It is here that hardware-in-the-loop (HIL) simulation emerges as a solution to mitigate these issues. HIL simulation addresses these inaccuracies through the integration of physical hardware such as actuators and PTO systems. The inclusion of physical hardware in the loop substantially reduces room for errors and uncertainties. Additionally, this integration enables a direct interface between computational models and physical components, thus refining the fidelity of simulations and reducing the gap between theoretical models and actual performance.

As marine hydrokinetic (MHK) energy, specifically ocean current turbines (OCTs), progresses toward commercialization, research is increasingly focused on improving turbine efficiency, reliability, and control. Key areas of recent research include advancing ocean environment models [10,12], refining dynamic simulations of turbines [13, 14], implementing advanced control methods [15–17], and developing innovative approaches for assessing turbine reliability and maintenance [18–20]. Hardware-in-the-loop (HIL) simulation, an established technique in wind turbine assessment, offers a promising method for OCT research by facilitating efficient performance testing and control system verification [21–23]. A well-designed HIL testbed can enhance experimental testing and validation, reducing costs while improving the development and deployment of OCTs.

HIL testing for MHK energy converters is a relatively new field, with limited existing literature. However, the similarities between OCTs and wind turbines provide valuable insights from wind turbine HIL studies. For instance, a power hardware-in-the-loop (PHIL) testbed was developed for utility-scale wind turbines to efficiently investigate power controls and low-voltage ride-through capabilities [24]. Additionally, HIL testbeds have been used to verify control methods before wind turbine deployment, demonstrating cost-effectiveness and reliability [25]. More sophisticated systems, integrating wind tunnels into the testbed, reduce the computational demands of aerodynamic modeling [22]. While most HIL literature focuses on wind turbines, studies have adapted similar methods for MHK systems, such as simulating marine current turbines with turbine-generator setups, though many omit critical factors like hydrodynamic effects [26]. Grid-integrated

PHIL testing is even less explored for OCTs, but the success of PHIL in wind turbine grid compatibility, fault testing, and power system stability [27,28], along with integration of alternative power sources such as water electrolysis [29], highlights the potential of PHIL simulations to advance OCT control systems and grid integration.

The motivation for developing and utilizing a HIL testbed for OCTs stems from the need to bridge the gap between purely simulated environments and real-world applications. While traditional simulations, such as those conducted in Simulink, offer valuable theoretical insights, they often fall short in capturing the complex interactions and dynamic behaviors of physical systems. The HIL approach addresses this limitation by integrating actual hardware components into the simulation loop, allowing for real-time testing under conditions that closely mimic real-world scenarios. This capability is crucial for accurately assessing control strategies and system performance in a way that purely simulated models cannot.

This paper explores the modeling, simulation, and hardware-in-the-loop testing of OCTs within a grid-connected framework. It begins with the system modeling and simulation setup and moves into the HIL implementation and description. Five case studies were then conducted to (1) analyze the OCT's torque response under extreme weather conditions, (2) evaluate a blade pitch controller in a real-time simulated environment, (3) examine the validity of the HIL feedback, (4) analyze a variable-speed controller in a HIL environment, and (5) analyze both the variable-speed controller and blade pitch controller operating in region 2.5.

2. System modeling and simulation setup

Before a HIL platform can be designed for an OCT, a detailed numerical system model is essential. This OCT model must first undergo an initial simulation to observe its performance in realistic ocean current conditions. This simulation will then be followed by incorporating a microgrid to observe grid-integrated behaviors. This section presents the numerical simulation design and setup.

2.1. Ocean environment

To simulate the ocean environment, either a constant ocean flow speed or Acoustic Doppler Current Profiler (ADCP) water velocity data are used as a base flow velocity and are then subject to modification to enhance realism, accounting for factors including wave/structure interactions defined using the significant wave height (H_s), turbulence impacts using the water turbulence intensity (TI), and the depth of the rotor center (d), as specified in [10].

The total water velocity, U_{tot} , is found after calculating the velocity due to waves and turbulence [30], and is given by:

$$U_{tot} = U + U_{wave} + U_{turb} \quad (1)$$

where U can represent either lowpass filtered ADCP data or a constant value.

The wave-induced water velocity, denoted as U_{wave} , is computed as a function of H_s and d using the following wave spectrum. This spectrum is converted into the time domain using a discrete sampling interval, n . Specifically, U_{wave} is represented by the following equation:

$$U_{wave} = \sum_{i=1}^n \left(A_i w_i \frac{\cosh(k_i(h-d))}{\sinh(k_i) \cdot h} \cdot \cos(w_i \cdot t_i + \phi_i) \right) \quad (2)$$

where w is the wave frequency, $k = \frac{2\pi}{w} \cdot \frac{2\pi}{h}$ is the wave number, h is the total water depth (i.e., 320 m for this study), ϕ is a random phase assigned to each wave frequency component, and A is the wave amplitude derived from the following equation where $\alpha = 0.0081g^2$ and $\beta = \frac{3.11}{H_s^2}$:

$$A = \sqrt{2\alpha w^{-5} e^{\beta w^{-4}} \cdot \frac{w_{max} - w_{min}}{n-1}} \quad (3)$$

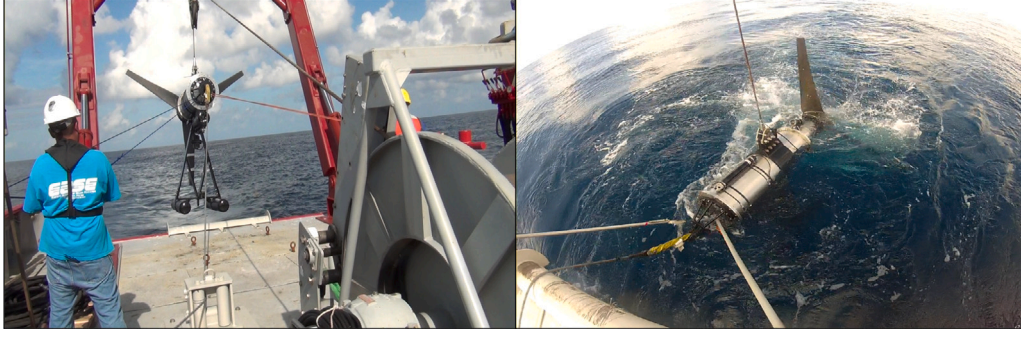


Fig. 1. FAU prototype OCT offshore testing in the Gulf Stream.

The turbulence spectra, denoted as T_{spec} , is calculated as a function of TI , mean water velocity magnitude U , and the considered turbulence frequency range w_{turb} , discretized as $f = \frac{w_{turb}}{2\pi}$, where an $f^{-\frac{5}{3}}$ decay is assumed over this range. The equation is given by:

$$T_{spec} = 2 \cdot df \cdot \left(\frac{\frac{2}{3} \cdot TI^2 \cdot U^2 \cdot f^{-\frac{5}{3}}}{\left(\frac{1}{(f_{min})^{2/3}} - \frac{1}{(f_{max})^{2/3}} \right)} \right)^{0.5} \quad (4)$$

where the discrete step size $df = 1.5915e^{-4}$. From T_{spec} , the flow speed due to turbulence is calculated as:

$$U_{turb} = \sum_{i=1}^l T_{spec_i} \cdot \cos(w_{turb_i} \cdot t_i + \phi_i) \quad (5)$$

where l is the length of T_{spec} .

2.2. Ocean current turbine modeling

The OCT rotor blade model presented in [17] was created as a variable pitch rotor blade option to match the experimental OCT that was deployed and tested in the Gulf Stream, shown in Fig. 1. The OCT simulation calculates mechanical rotor torque, T_m , based on flow speed, U , rotor radius, r , low-speed shaft speed, s , and blade pitch angle. T_m begins with the conversion of s to tip-speed-ratio, TSR , as follows:

$$TSR = \frac{2\pi rs}{60U} \quad (6)$$

For case studies 1 and 3 blade pitch control is not used. Therefore, blade pitch root angles are constant and the torque coefficient, C_τ , is determined from methods following [31,32], where real-time C_τ is estimated through polynomial regression using WT_Perf for incremental TSR values. The estimated C_τ accommodates the discontinuity between R_0 and R_1 through a linear interpolation:

$$C_\tau^e = \begin{cases} C_\tau^{R_0}(TSR), & \text{if } TSR \leq 3.45 \\ C_\tau^{R_1}(TSR), & \text{if } TSR \geq 3.61 \\ (C_\tau^{R_1}(3.61) - C_\tau^{R_0}(3.45)) \frac{TSR - 3.45}{0.16}, & \text{otherwise} \end{cases} \quad (7)$$

where R_0 and R_1 define C_τ as a function of TSR as:

$$C_\tau^{R_0} = 0.00974 + 0.00958 \cdot TSR + 0.01403 \cdot TSR^2 - 0.00216 \cdot TSR^3, R_0 \in 0 - 3.45 \quad (8)$$

$$C_\tau^{R_1} = 0.21562 + 0.03370 \cdot TSR + 0.00218 \cdot TSR^2 - 0.00007 \cdot TSR^3, R_1 \in 3.61 - 14.14 \quad (9)$$

In cases 2 and 4, variable blade pitch control is utilized and therefore must be considered when calculating rotor performance. To do this a family of torque and power coefficient curves were derived for a range of blade pitch root angles using the same 20 m rotor model [17].

These data were integrated into a 2-D look-up table in Simulink to generate interpolated torque curves, while a separate 1-D look-up table is employed for the power coefficient curves being independent of blade pitch. The torque coefficient curve is illustrated in Fig. 4 and was computed for blade pitch angles ranging from -10° to 10° with 2° increments where the torque coefficient C_τ and power coefficient C_p are defined as shown below:

$$C_p = \frac{P}{0.5\rho A_s(U)^3} \quad (10)$$

$$C_\tau = \frac{T}{0.5\rho A_s(U)^2} \quad (11)$$

where P is the predicted shaft power, T is the predicted shaft torque, ρ represents the assumed seawater density (1025 kg/m^3), and A_s is the rotor's swept area (314.16 m^2) which is based off of the 20 m diameter rotor model previously mentioned.

For all cases, the final output is then calculated as:

$$T_m = 0.5\rho A_s U^2 r C_\tau^e \quad (12)$$

where T_m is then converted from Nm to per unit ($p.u.$) for MATLAB simulation [33], considering the asynchronous generator's nominal power rating P_{nom} :

$$p.u. = \frac{Nm}{P_{nom} \text{ kVA}} \quad (13)$$

2.3. Microgrid modeling

The microgrid model presented in this paper, shown in Fig. 2, is derived from the "Wind-Turbine Asynchronous Generator in Isolated Network" Matlab demo [34], originally developed by Hydro-Quebec. It represents a high wind penetration, wind-diesel system without storage, and has been adapted for use within the RT-Lab environment as an OCT-diesel system where the wind turbine is replaced with the OCT [35].

The system represents a high renewable energy penetration microgrid, and it simulates the performance of an ocean current turbine-diesel hybrid system without energy storage. The microgrid includes a 50 kW consumer load and a variable secondary load ranging from 0 to 446.25 kW to absorb any surplus power generated by the OCT.

The microgrid operates with a 480 V, 275 kVA asynchronous generator, supported by a 480 V, 300 kVA diesel-driven synchronous machine. The synchronous machine fulfills two key roles: during periods of low current speeds, it ensures sufficient power supply to the consumer load, and during periods of over-generation, it acts as a synchronous condenser, stabilizing reactive power within the system. To maintain the grid's stability, a phase-locked loop (PLL) system ensures a consistent grid frequency of 60 Hz by comparing the actual system frequency with a reference signal.

Additionally, the OCT power output line interacts with both the main consumer load and the variable secondary load, with the latter

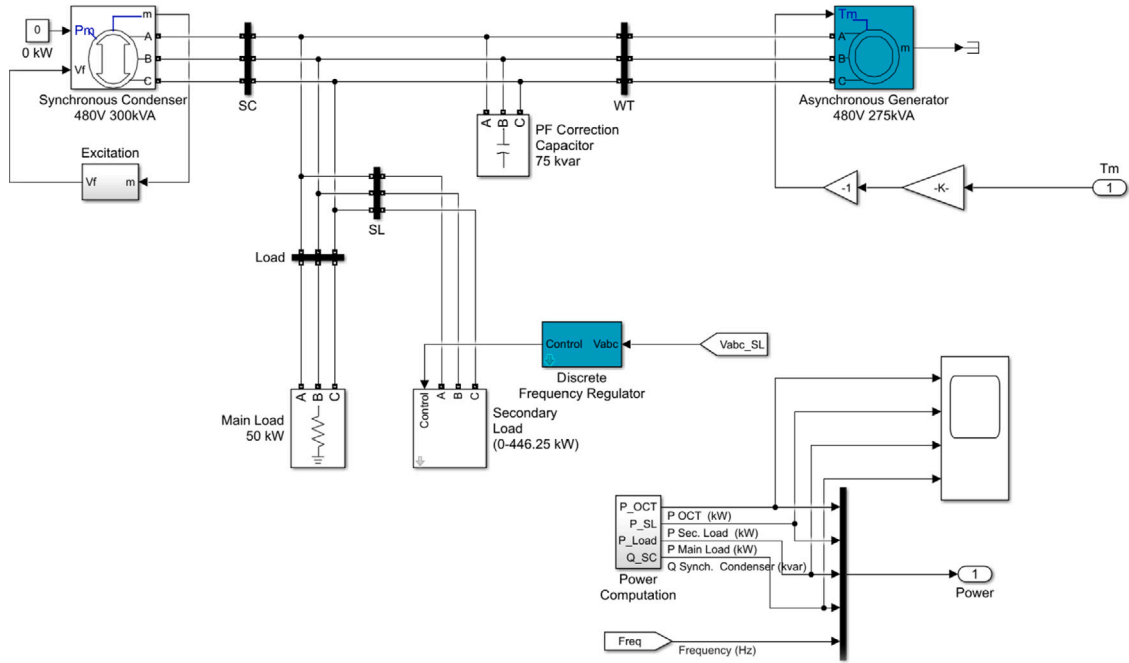


Fig. 2. Asynchronous generator connected to the microgrid with a mechanical torque input (T_m) coming from an OCT source (e.g., simulation or dynamometer).

absorbing any excess generation from the OCT. During times when OCT power is insufficient, the synchronous machine compensates by injecting power to maintain balance. This relationship between the OCT power, consumer load, and secondary load is critical for ensuring the smooth operation of the microgrid under varying ocean current speeds.

2.4. Controller modeling

This subsection outlines the control methodologies employed across the different regions of power generation. In Region 3, the blade-pitch controller is used exclusively to regulate power, ensuring it remains at or below the rated generation capacity. In Region 2.5, both the variable-speed and blade-pitch controllers work in tandem to facilitate a smooth transition between regions, minimizing overshoot and ensuring safe operation. In Region 2, the variable-speed controller alone is employed to maximize the power coefficient, C_p , functioning as a rudimentary Maximum Power Point Tracking (MPPT) controller. Fig. 3 illustrates the closed-loop power curve, defining each region. The cut-in current velocity for Region 2 is set at 0.8 m/s, and the rated current velocity is at 1.8 m/s. Region 2.5 is characterized by the generator operating at 80%–95% of its rated power, corresponding to a current velocity range from 1.57 m/s to 1.74 m/s. Both PID controllers were tuned manually using the basic trial-and-error method and the parameters remain constant across all regions.

2.4.1. Region 3 control

In Region 3 control, as outlined in [36], the blade pitch controller is used to prevent the turbine's power production from exceeding the rated capacity of the generator. To maintain power production at the designated 275 kVA, the turbine's blade pitch is dynamically adjusted, effectively reducing the angle of attack on the turbine and subsequently decreasing torque on the generator.

The design of this controller is inspired by [37,38] and utilizes a Proportional-Integral (PI) controller similar to [39]. The availability of the calculated torque coefficient curve for this blade model allows for the creation of a 2-D lookup table. This table is what enables the

controller to effectively manipulate the blade pitch of the OCT. The controller uses power generation feedback as its input and adjusts the blade-pitch angle to ensure it remains within a safe range for generator safety. When the system begins to transition into region 3, the blade pitch angle undergoes adjustment using a manually tuned PI controller, as depicted in Fig. 4.

2.4.2. Region 2 control

In Region 2 of power generation, the focus shifts from limiting power generation to optimizing power capture. This is commonly achieved through variable-speed control, a method which is extensively applied in wind turbines as shown in [40] and is equally effective for OCTs. This approach dynamically adjusts the turbine's rotor speed to maximize the power coefficient (C_p) by achieving the optimal TSR.

The power coefficient (C_p) indicates the efficiency of power capture and is dependent on the blade pitch and the TSR as analytically described in detail in [41]. In this region, a monovariable control strategy is used and therefore the blade pitch is fixed at 0° within the Region 2 mode of operation. As previously discussed, the power coefficient curve is derived from the interpolation of simulated data-points. With a constant blade pitch of 0° the observed C_p curve peaks at a value of 0.5679 occurring at a TSR of 10.13 and serving as a benchmark for optimal performance and a reference for the controller.

Many control methodologies have been proven effective for managing variable speed, with the sliding mode controller, detailed in [42], being one notable example. The controller chosen for this experiment is a simple manually tuned PI controller which has been extensively used for variable-speed control on wind turbines [43]. To accurately track and adjust to the maximum C_p , a 1-D lookup table of the power coefficient curve was incorporated into the Simulink model. This table takes TSR as an input, which is calculated from the RPM feedback of the dynamometer and ocean environmental model flow speed, to calculate the measured power coefficient. A PI controller takes the error from the measured C_p and adjusts the RPM input to the dynamometer, as shown in Fig. 5. Given the measured C_p the torque coefficient can be calculated using $C_t^e(\lambda, \beta) = \frac{C_p(\lambda, \beta)}{\lambda}$. Additionally, a simple switch

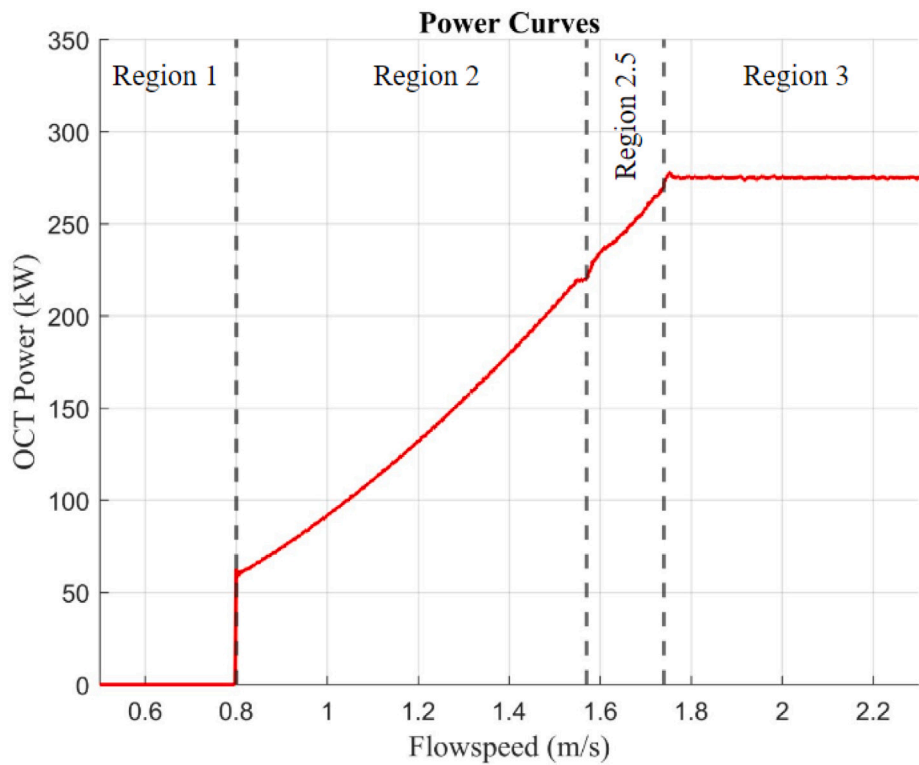


Fig. 3. Power curves defining regions 2, 2.5, and 3.

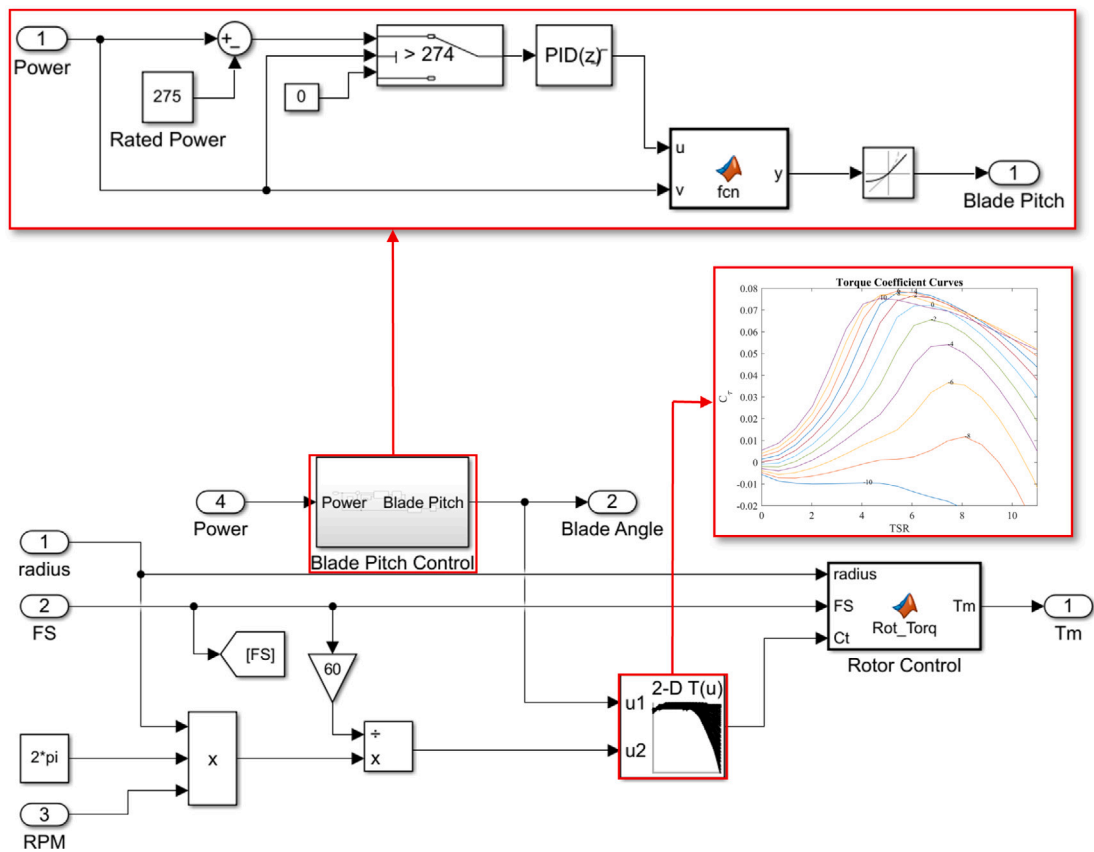


Fig. 4. Blade pitch controller Simulink model with torque coefficient curves look-up table.

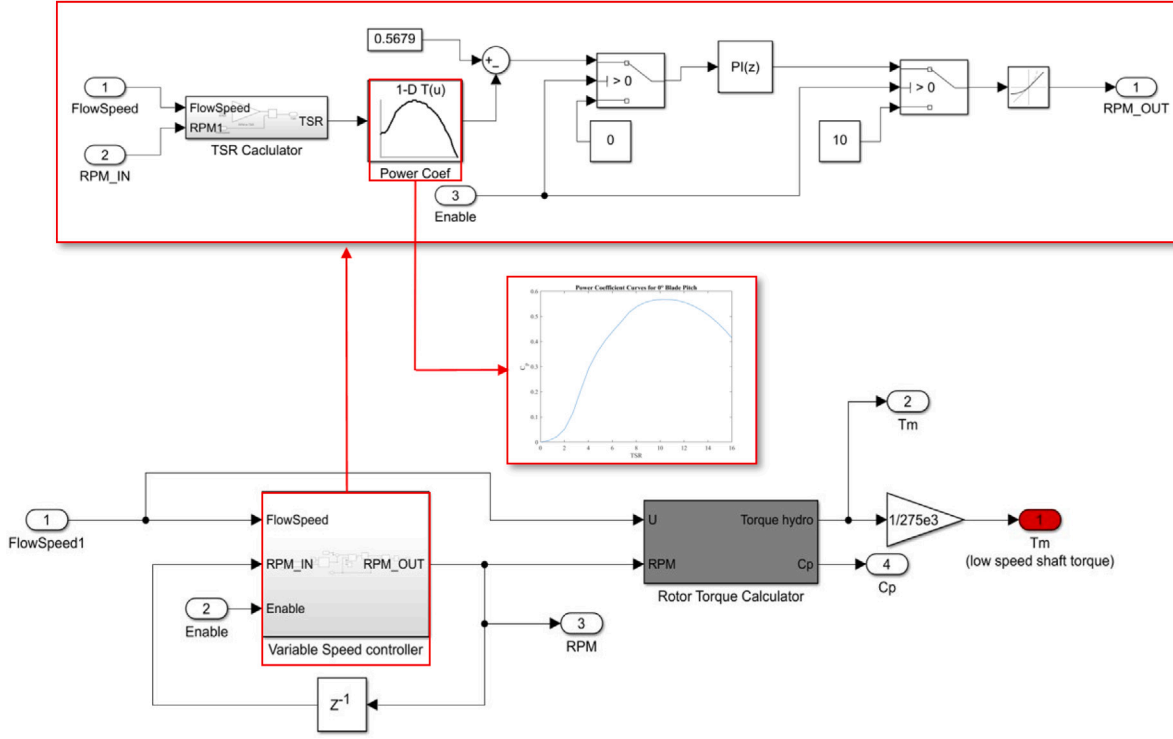


Fig. 5. Variable speed controller Simulink model with power coefficient curve look-up table.

mechanism is used as an enable the PI controller to prevent integral wind-up when the RPM is being controlled by the user.

2.4.3. Region 2.5 control

In Region 2.5, a hybrid control strategy combines the variable-speed control from Region 2 and the blade-pitch control from Region 3 to ensure a smooth and efficient transition as the system approaches Region 3, as discussed in [44]. In this region, where the flow speed nears the rated value, both controllers use a basic dynamic reference algorithm to adjust the PI reference points, ensuring a seamless transition between power maximization and power limitation modes. For simplicity in this region, the reference point for both controllers are defined by linear Eqs. (14) and (15) which depend on the measured flow speed (U). In these equations, P_{rated} is 275 kVA, $C_{p,max}$ is 0.5679 (as discussed in the previous section), and $C_{p,rated}$ corresponds to the desired TSR at the cut-out current velocity, set at 0.33 in this scenario.

$$P_{ref} = 0.8P_{rated} + \left(\frac{U - 1.57}{1.74 - 1.57}\right)(0.95P_{rated} - 0.8P_{rated}) \quad (14)$$

$$C_{p,ref} = C_{p,max} + \left(\frac{U - 1.57}{1.74 - 1.57}\right)(C_{p,rated} - C_{p,max}) \quad (15)$$

The dynamic reference algorithm causes the blade-pitch controller to gradually engage at a flow speed of 1.57 m/s, enabling a smooth transition as the turbine shifts from Region 2 to Region 3. Simultaneously, the variable-speed controller adjusts the rotor speed towards the constant-speed target required for Region 3. While more sophisticated dynamic reference or reference-bias control algorithms, such as those discussed in [45], offer improved performance, this study opted for a simpler, linear approach. Although less optimal, the linear method was chosen to align with the scope of this paper.

3. Hardware description and implementation

3.1. Dynamometer specifications

The dynamometer test-bed consists of two AC induction motors mechanically coupled together with zero backlash couplings and a Lebow 1604 torque cell positioned between them for torque measurements. The setup includes a prime mover, which is a 2 HP, 1800 RPM induction motor, and a motor under test (MUT), which is a 3 HP, 1800 RPM inverter-duty motor. Both motors are controlled via standard off-the-shelf Variable Frequency Drives (VFDs), which was shown to be an effective method [46]. The prime mover's VFD is set to be torque controlled allowing it to maintain a steady torque based on the reference value given, while the MUT VFD is set to speed control to act as an active generator load. The MUT is equipped with dual quadrature encoders to facilitate speed feedback to both the VFD and the Opal-RT, allowing the ability for a closed-loop system. The VFDs output the torque information back to the Opal-RT via analog signaling, which is necessary to drive the generator in the Simulink microgrid model. A full depiction of the lab setup and communication flow chart can be seen in Fig. 6. This dynamometer test-bed is used to emulate the rotor in this setup and leverages the mechanical feedback to drive the simulated generator.

3.2. Hardware-in-the-loop integration

The Opal-RT 4510 utilizes RT-Labs software for its real-time simulations. The RT-Labs environment is capable of integrating with MATLAB/Simulink models, which is beneficial when designing dynamic and control systems models. This compatibility facilitates more efficient designs of real-time models and complex systems.

Creating a HIL test-bed involves integrating the Opal-RT to work as a closed-loop system with the dynamometer. The integration process

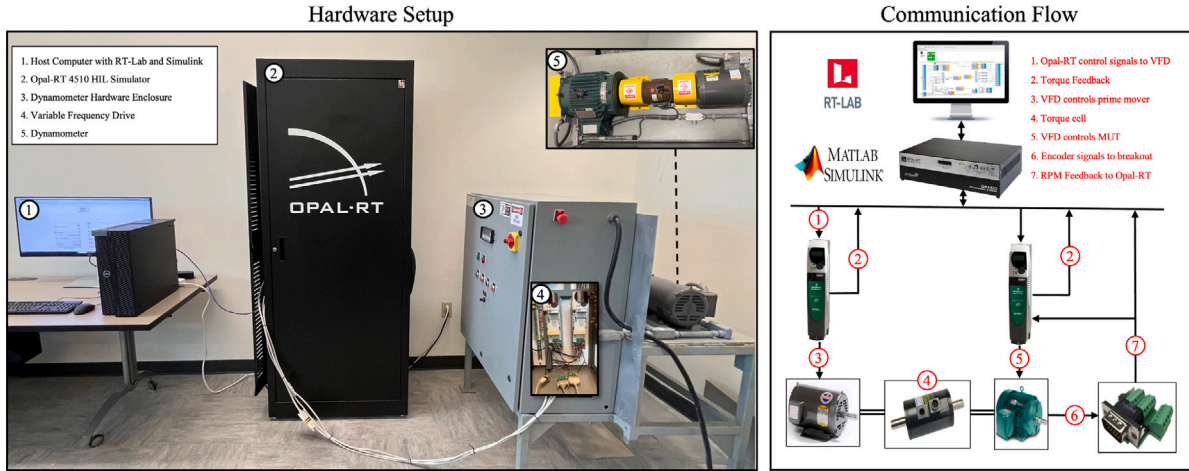


Fig. 6. Hardware-in-the-loop lab setup (left) and communication flowchart (right).

focuses on configuring the Opal-RT to communicate effectively with the dynamometer's VFDs in order to regulate the motors' speed and torque. This communication is facilitated through the Opal-RT's analog and digital I/O ports, allowing the transfer of control signals between the simulator and the test-bed. In this setup, the Opal-RT generates and sends normalized 0–10 V analog signals to the VFDs which dictates the torque for the prime mover and the speed for the MUT. Additionally, the Opal-RT sends digital signals to the VFDs to control the run of the motors such as drive enable and run forward, as dictated by the Simulink model.

As the Opal-RT sends signals to the dynamometer, the dynamometer also sends signals back to the Opal. As previously stated, the MUT contains dual quadrature encoders that measure the RPM of the motor. The Opal-RT 4510 features two dedicated encoder inputs which enables simple feedback interpretation. The encoder ports read the RPM of the motor shaft and utilizes this along with the user-input flow speed to calculate the TSR of the system which is essential to determining the rotor's torque coefficient characteristics.

This integrated system enables a high degree of control and precision in testing scenarios, as the Opal-RT can simulate a wide range of operational conditions and feed this information into the test-bed in real time. The dynamometer adjusts the motors' performance according to the modeled environment, providing valuable data on how the turbine generator system behaves under various simulated conditions. This integration is beneficial for testing and validating control strategies, system responses, and the durability of components under different load conditions.

3.3. Rotor model scaling

The rotor model in this study, which is based on the 20 m rotor model from [17], requires adaptation to fit the dynamometer's physical constraints. While retaining the original blade model design, the rotor's radius is adjusted to better match the dynamometer. By reducing the radius of the rotor model and increasing the RPM of the dynamometer, TSR specifications are maintained.

The revised rotor model's radius is calculated based on equations from [47] to support a maximum power generation P_{gen} of 1.6 kW. The calculation uses the water density ρ (1025 kg/m^3), a maximum power coefficient C_p (assumed to be Betz limit of 0.59), and a target maximum water velocity V_o of 3 m/s: $r = \sqrt{\frac{2P_{gen}}{C_p \rho V_o^3 \pi}}$. Using the equation, a rotor

diameter of 0.5 meters is calculated allowing for the implementation of an effective rotor model. Due to consistent TSR characteristics related to the rotor blade design and linear scaling, the torque and power coefficient curves remain valid for scaled-down models, allowing the use of the same variable-speed and blade-pitch controllers.

As previously discussed, the microgrid model in this study utilizes a 275 kVA generator which is based on a 20-meter OCT being its prime mover. Since the dynamometer in this study is not equipped to produce the amount of torque needed to drive this generator, a torque scaling method was implemented through TSR matching between the 20 m rotor and 0.5 m rotor, where the hydrodynamic performance characteristics are a close match between both models. Given both the 20 m rotor model and the 0.5 m rotor model, a torque scaling factor can be calculated between the models, and the relationship can be graphically shown in Fig. 7.

4. Case study

This section outlines the five case scenarios and presents the simulation results of each.

4.1. Case 1: OCT mechanical torque simulation in hurricane conditions

The mechanical torque of the OCT operating at a depth of $d = 20\text{m}$ for a $r = 10\text{m}$ rotor is modeled operating at $s = 10$ RPM using 1000 s of ADCP water velocity data as a flow speed input. The dataset was obtained from the Gulf Stream at a latitude of 26.094° N and a longitude of -79.800° E on September 11, 2017, during Hurricane Irma. Different turbulence intensities (TI) are applied to the data to analyze variations in fluctuations, and the outcomes are depicted in Fig. 8.

The ADCP data is modified as described in 2.1 with the parameter T_{spec} calculated as a function of $TI = 0.15$, $U = 1.16 \text{ m/s}$, and $w_{turb} = 0.1 - 2\pi \text{ rad/s}$, discretized as $f = \frac{w_{turb}}{2\pi}$. The significant wave height is set to 5 m. This wave-height is based on data presented in a national report [48] which analyzes the ocean conditions at various locations during Hurricane Irma. At 5% TI, the maximum torque output is 922,216 Nm and the minimum is 285,836 Nm. Similarly, at 30% TI, the maximum is 995,097 Nm and the minimum is 181,833 Nm, and observation shows that a 25% increase in TI leads to much greater fluctuation amplitude and frequency.

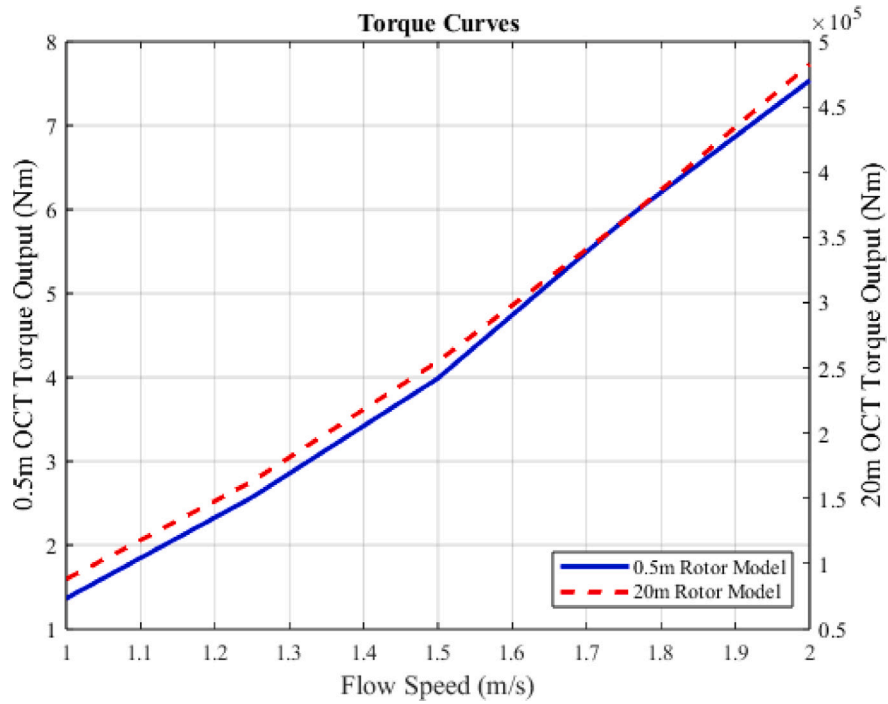


Fig. 7. Torque scaling between the 20 m rotor diameter and 0.5 m rotor diameter at increasing flow speeds.

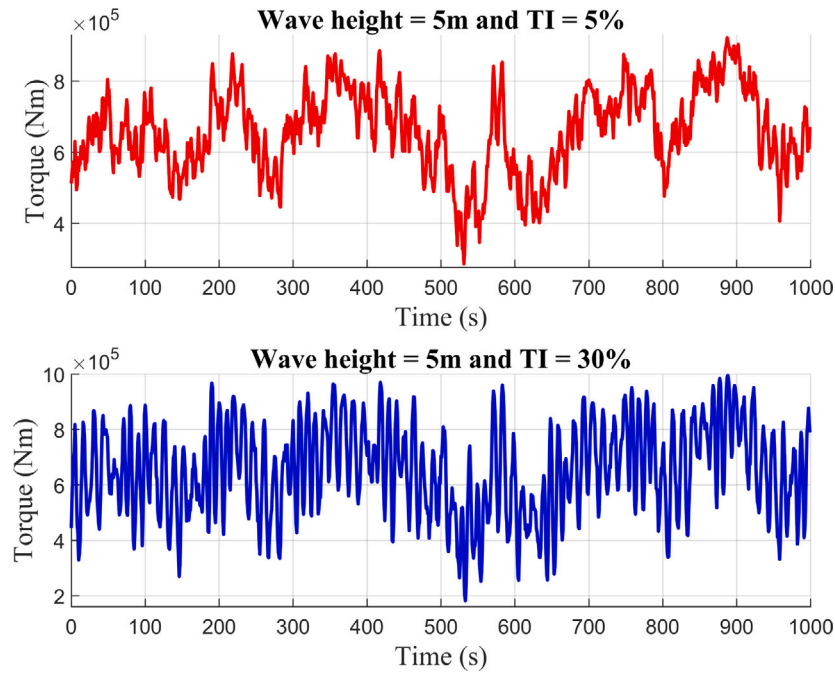


Fig. 8. OCT mechanical torque output with 1000s hurricane conditions ADCP data input at 5% and 30% TI.

4.2. Case 2: Ocean current flow speed step change with blade pitch control

In this case study, the OCT model, linked to the microgrid described in 2.3, operates with an OCT depth of 50 m. Ocean current step changes are applied at 5s to evaluate grid frequency and controller performance.

After the step change, the system enters Region 3 where the blade pitch controller activates to reduce the amount of power generated by the OCT. First, a step change from 1.4 m/s to 1.6 m/s is applied and the blade pitch controller activates and the blade pitch is observed to peak

at an angle of 4.5° and then reaches a steady-state of 3.6° as shown in Fig. 9. Then a step change from 1.4 m/s to 1.8 m/s is applied and the blade pitch controller is observed to reach a steady-state angle of 10.5° . In both step change cases, the grid frequency is seen to oscillate when the step change is introduced and settles back to the desired value of 60 Hz at around 16 s.

Fig. 10 shows the microgrid response with the blade-pitch controller implemented. At both step changes it is shown that the blade-pitch regulates the OCT power generation down to 275 kW. The step change

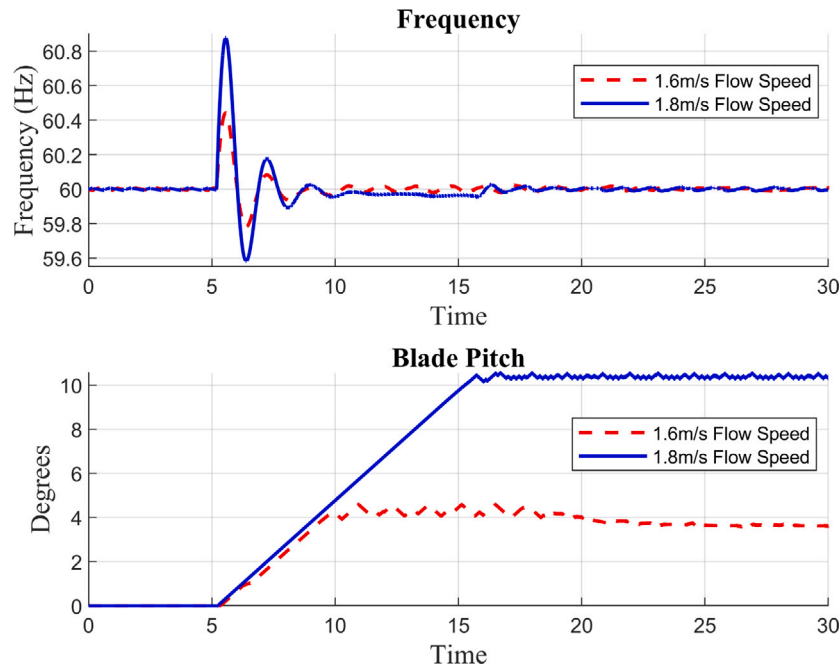


Fig. 9. Grid frequency and blade pitch controller performance during an ocean current input step change of 1.4 m/s to 1.6 m/s and 1.4 m/s to 1.8 m/s.

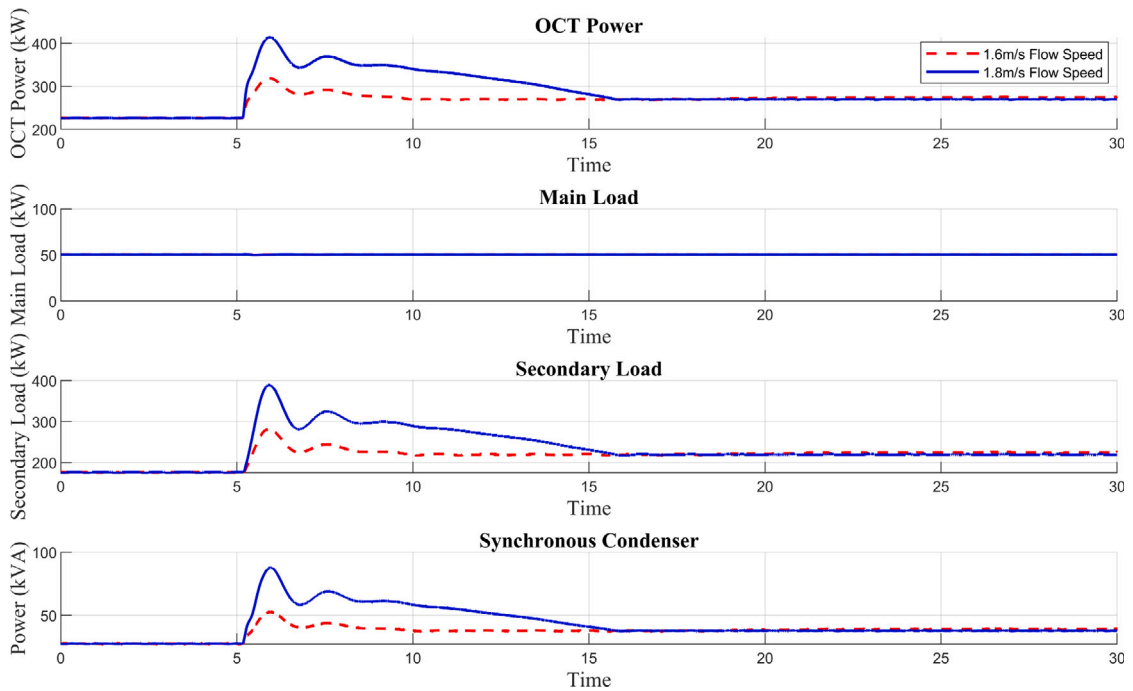


Fig. 10. Microgrid response with blade-pitch controller activation and flow speed step changes of 1.4 m/s to 1.6 m/s and 1.4 m/s to 1.8 m/s.

to 1.6 m/s caused the power generation to spike to 315 kW and regulate back down to 275 kW, and the 1.8 m/s step change caused it to spike to 410 kW.

4.3. Case 3: OCT HIL feedback validation

This section of the study focuses on validating the HIL setup's capability to receive accurate real-time feedback corresponding to the torque inputs generated by the dynamometer under a range of operational conditions. To assess this, a series of experiments were conducted where the OCT is simulated at a depth of 1 m and the TI is incrementally adjusted and tested at thresholds of 5%, 10%, and 15%. These

tests were performed under significant wave height conditions of 0 m (Sea State 0) and 1 m (Sea State 4) [49], to simulate environments from calm to moderate sea states. The outcomes of these experiments are illustrated in Fig. 11, showcasing the system's ability to maintain stable feedback during varying degrees of environmental fluctuations, corroborating the validity of the test-bed operation.

At a wave height of 0 m, the increasing TI is seen to cause larger differences in torque output, with TI=5% having a range from 1.43 Nm to 1.98 Nm, TI=10% having a range from 1.19 Nm to 2.26 Nm, and TI=15% having a range from 0.97 Nm to 2.53 Nm. A wave height of 1 m introduces larger oscillations at a smaller frequency which greatly affects the range and smoothness of the torque output. At a wave height

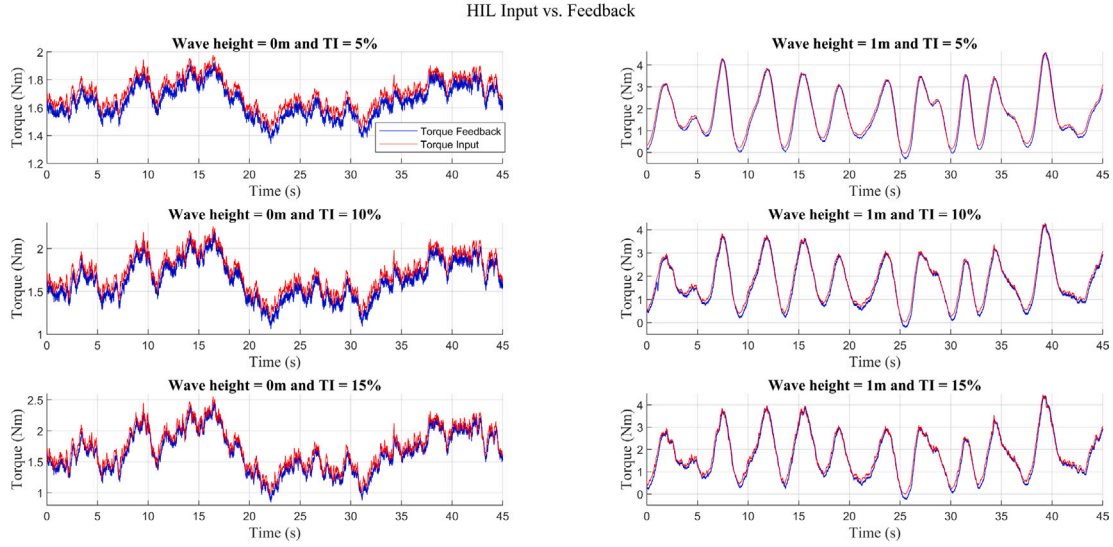


Fig. 11. HIL feedback validation varying from 5% to 15% TI at a significant wave height of 0 m (left) and 1 m (right).

of 1 m, the changes in TI fluctuations are primarily hidden in the high amplitude fluctuations that occur over a much narrower frequency band than turbulence. At TI=5% the torque ranges from -0.26 Nm to 4.59 Nm; at a TI=10% it ranges from -0.21 Nm to 4.28 Nm; and at a TI=15% it ranges from -0.22 Nm to 4.46 Nm. The negative values are caused by the prime-mover undershooting a small amount when the torque input approaches 0. The standard deviation for each set of data is shown in Fig. 12 along with the value of the deviation. It is notable that the deviation values change significantly with variations in wave-height and minimally with turbulence intensity. With a wave-height of 0 m and a TI of 5%, 10%, and 15% we can see the deviation varies from 0.109, to 0.219, to 0.326 respectively. As the wave-height increases to 1 m we can see the deviation increases to 1.067, 0.933, and 0.952 for respective TI values of 5%, 10%, and 15%.

4.4. Case 4: OCT HIL with variable speed control

After validating the feedback from the dynamometer, a grid-connected case was conducted to observe the variable speed controller response along with the grid's power and frequency response.

Step changes at the flow speed input were applied manually from 1.0 m/s to 1.2 m/s at 17 s, and from 1.2 m/s to 1.4 m/s at 59 s as shown in Fig. 13. At 17 s, the power coefficient, C_p , is seen to drop from the desired max of 0.5679 to 0.551 at 17 s and regain the desired max at 24 s. At 59 s, C_p drops to 0.557 and regains the desired max at around 65 s. Results show the capability of the controller to maintain the desired C_p via changing RPM inputs while running in the HIL configuration.

Fig. 14 shows the power profile of the microgrid with the variable speed control. The system is observed from an initial steady state with a 1.0 m/s flow speed. The power at the OCT at this time is about 81 kW. At 17 s a step change from 1.0 m/s to 1.2 m/s is introduced and the power at the OCT increases to 131 kW and then settles to 112 kW, while the secondary load increases to 88 kW and settles to 62 kW, and the synchronous condenser increases to 12 kVA and settles to 9 kVA to compensate for the extra power being produced. At 59 s the step change to 1.4 m/s causes the power at the OCT to increase to approximately 175 kW, while the secondary load increases to 133 kW and settles to 102 kW, and the synchronous condenser increases to 18 kVA and settles to 14 kVA.

The frequency response of the microgrid can be seen in Fig. 15. At each step change, the frequency oscillates and becomes sporadic for a short duration. During the first step-change, the grid takes 13 s for the

frequency to settle at 60 Hz, and in the second step-change, it takes 20 s for the system to settle.

4.5. Case 5: Region 2.5 analysis and turbine AEP

This section evaluates the effectiveness of the control strategies implemented in Region 2.5. These controllers ensure a smooth transition between Regions 2 and 3, minimizing overshoot while maintaining the safety of the generator, turbine, and grid. As discussed earlier, Region 2.5 employs both blade-pitch and variable-speed controllers, using dynamic reference points to facilitate the transition. A simulation was performed to analyze a step change in flow speed from 1.7 m/s to 1.75 m/s, right at the boundary of entering Region 3. As shown in Fig. 16, the blade-pitch controller is already active with the controller in Region 2.5, while it remains inactive without the transition controller. The step change occurs at approximately 16 s in both models. Although the frequency response shows only a slight improvement with the controller (oscillating between a peak of 60.125 Hz and a trough of 59.94 Hz), the impact on power overshoot is significant. As illustrated in Fig. 17, without the Region 2.5 controller, the generator experiences a power overshoot, peaking at around 341 kW before stabilizing, whereas with the controller, the peak is limited to 280 kW. This demonstrates the importance of implementing a Region 2.5 controller for turbine safety during transitions. While more advanced control strategies such as neural network-based controls [50], model predictive controls [51], and fuzzy logic controls [52] offer enhanced optimization and performance, they come with increased complexity.

To evaluate the control methodology's effectiveness, the closed-loop Annual Energy Production (AEP) of the turbine is calculated using Eq. (16). In this equation, $P_i(U_i)$ represents the measured closed-loop power curves as shown in Fig. 3, $f_i(U_i)$ denotes the current velocity distribution shown in Fig. 18, N_h is the number of hours in a year (8760 h), ΔU is the velocity bin size (0.1 m/s), and N_B is the number of velocity bins used. The ocean current velocity distribution, shown in Fig. 18 is derived from data collected at a 50-meter depth using a vessel-mounted ADCP aboard the research vessel Walton Smith. This data was gathered along the 27°N transect line and is discussed in detail in [53]. Previous studies, such as [6], have shown that the velocity distribution of the Gulf Stream typically follows a Gaussian distribution. Accordingly, $f_i(U_i)$ is modeled as a Gaussian distribution with a mean of 1.494 m/s and a standard deviation of 0.341 m/s.

$$AEP = N_h \sum_{i=1}^{N_B} P_i(U_i) f_i(U_i) \Delta U \quad (16)$$

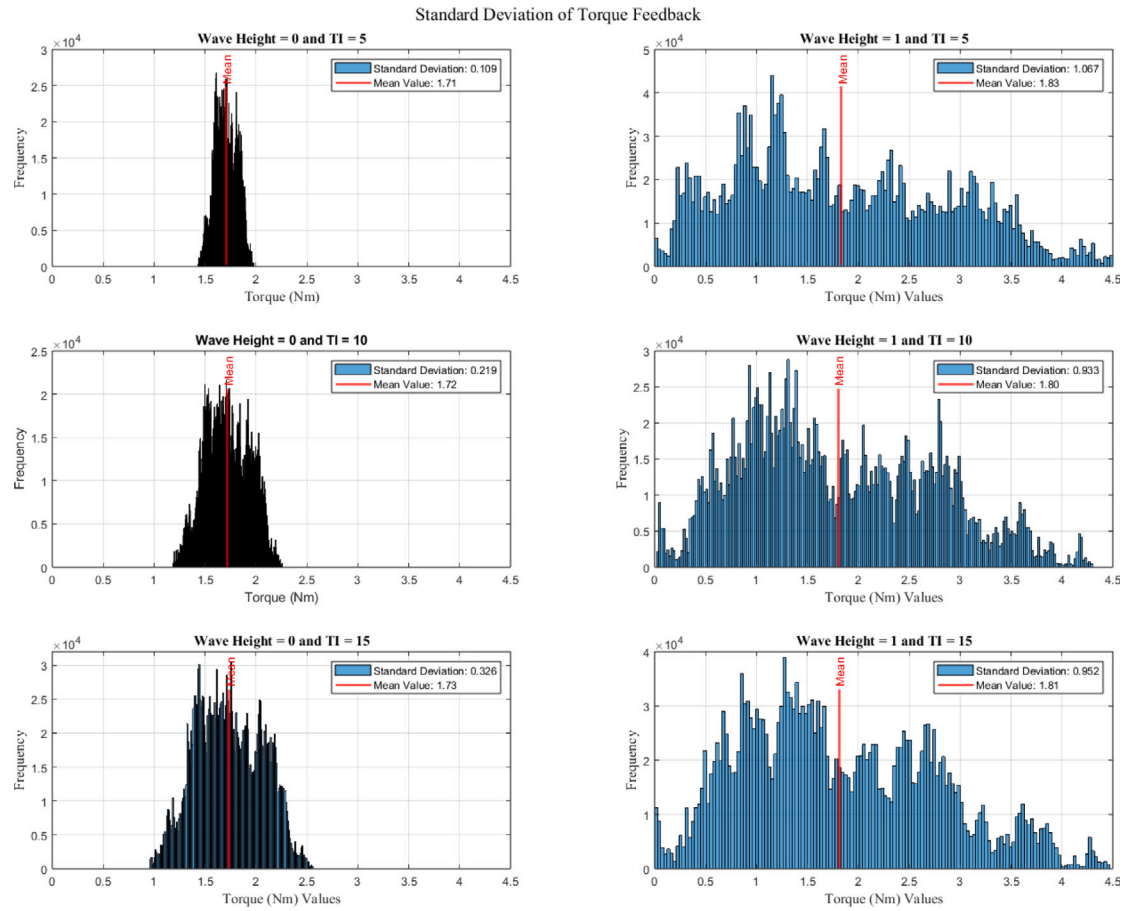


Fig. 12. Torque feedback deviation varying from 5% to 15% TI at a significant wave height of 0 m (left) and 1 m (right).

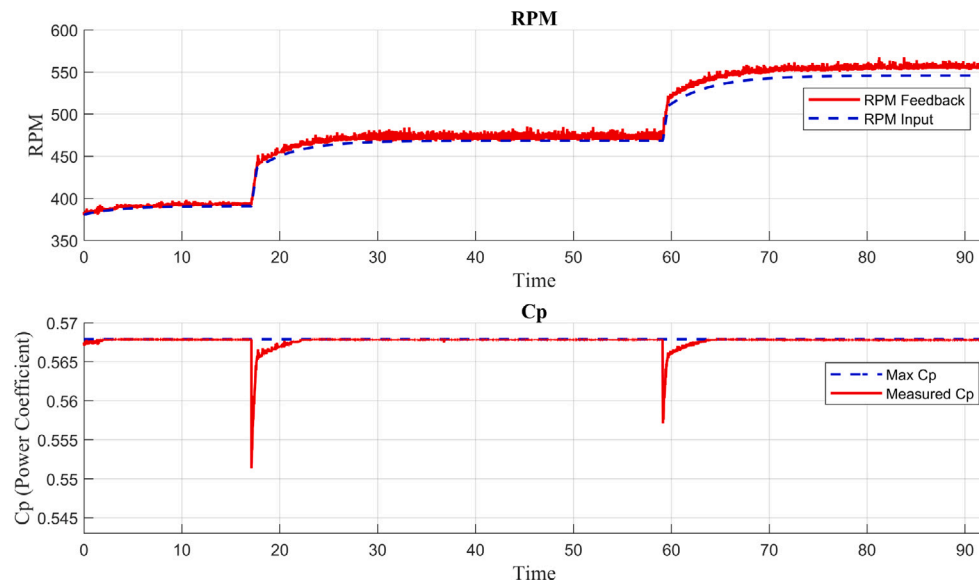


Fig. 13. Variable speed controller performance with RPM step changes from 1.0 m/s to 1.2 m/s and 1.2 m/s to 1.4 m/s (top), and power coefficient response (bottom).

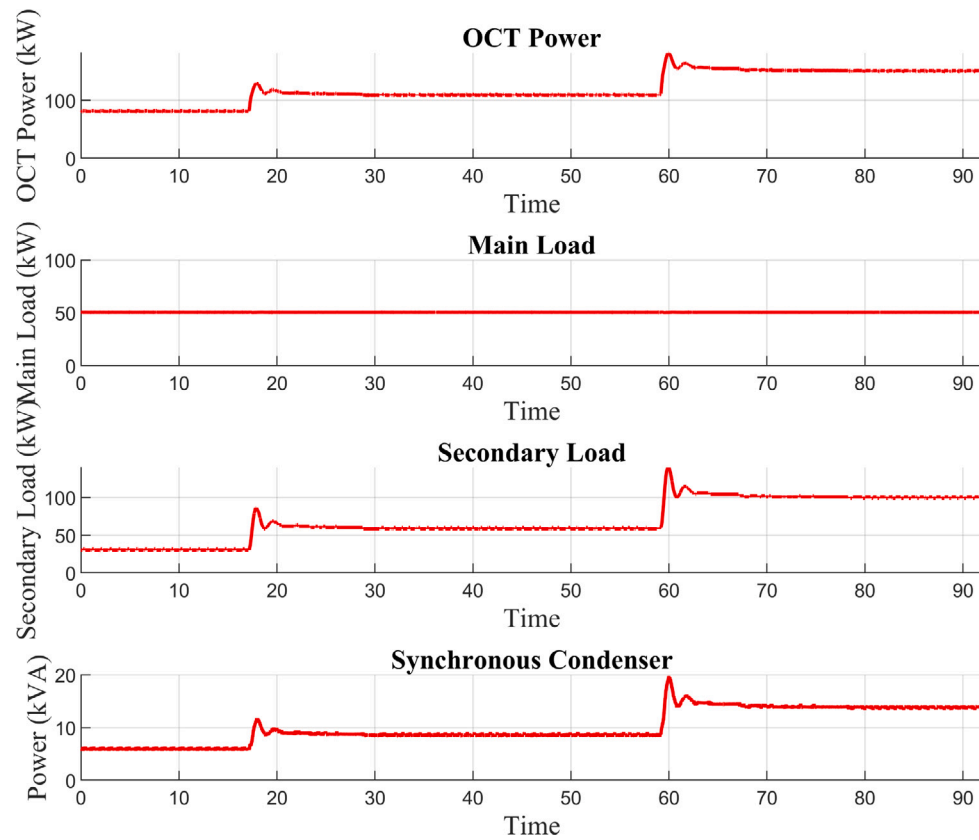


Fig. 14. The power from the OCT, main load, second load and synchronous condenser with flow speed input step changes from 1.0 m/s to 1.2 m/s and 1.2 m/s to 1.4 m/s under variable speed control.

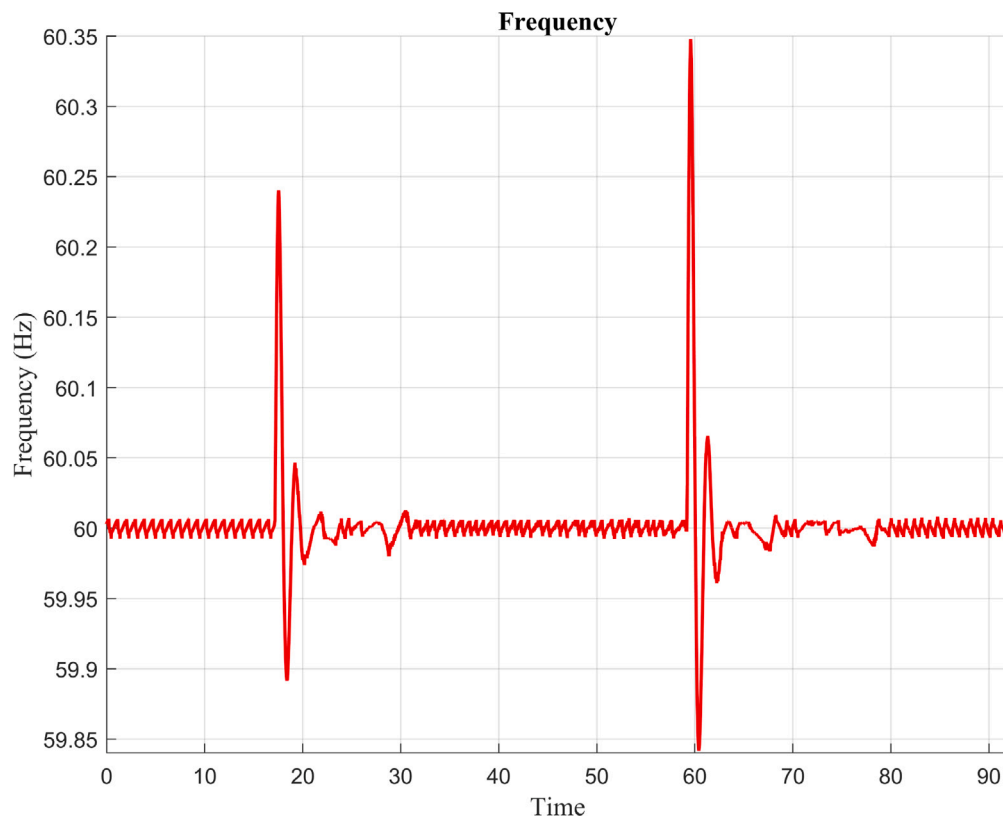


Fig. 15. Grid frequency response with flow speed input step changes from 1.0 m/s to 1.2 m/s and 1.2 m/s to 1.4 m/s under variable speed control.

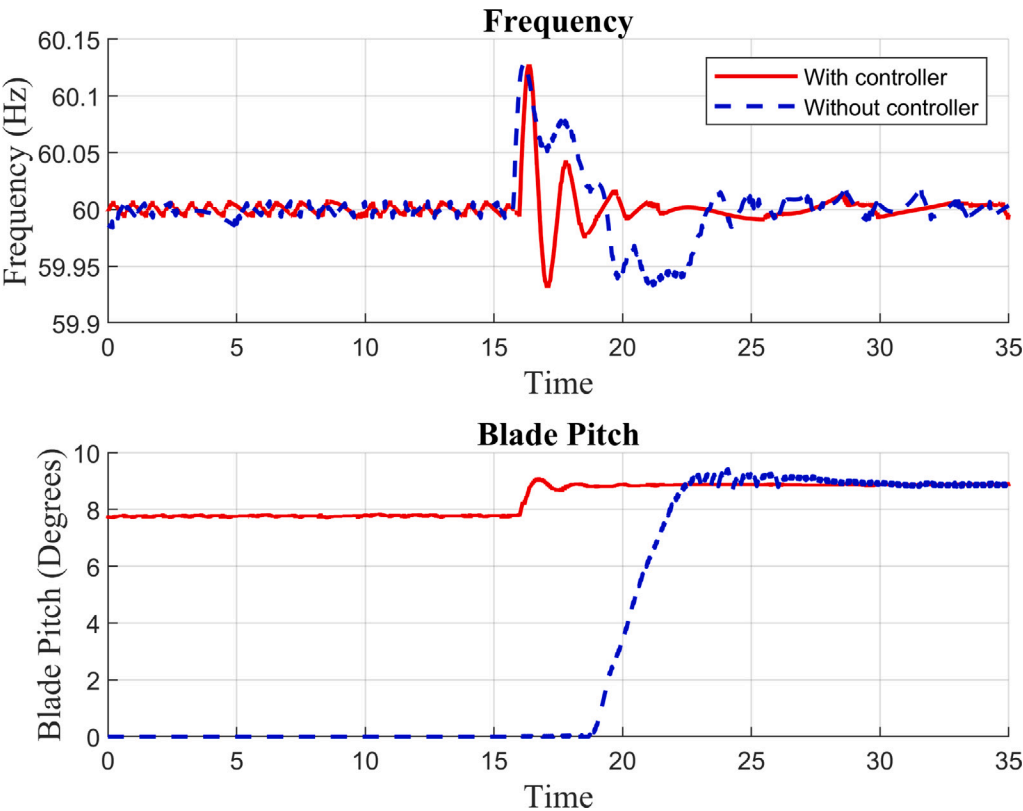


Fig. 16. Frequency and blade-pitch angle when changing from region 2.5 to 3 using a step change from 1.7 m/s to 1.75 m/s.

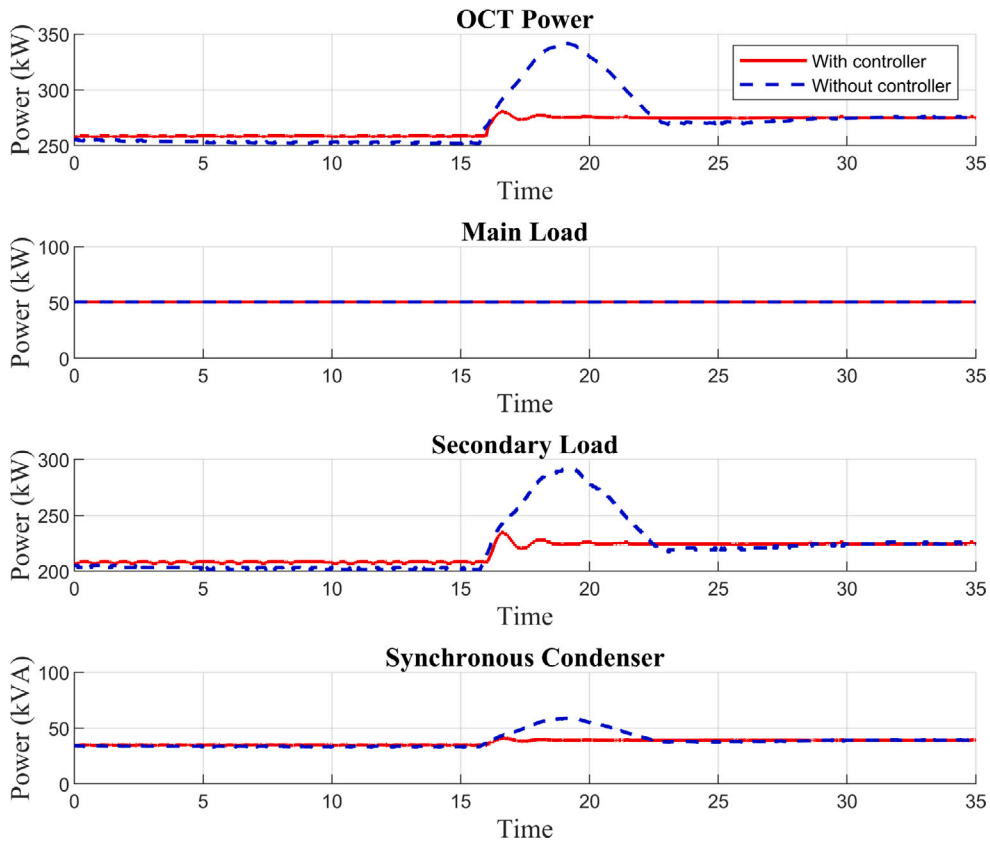


Fig. 17. Power from the OCT when changing from region 2.5 to 3 using a step change from 1.7 m/s to 1.75 m/s.

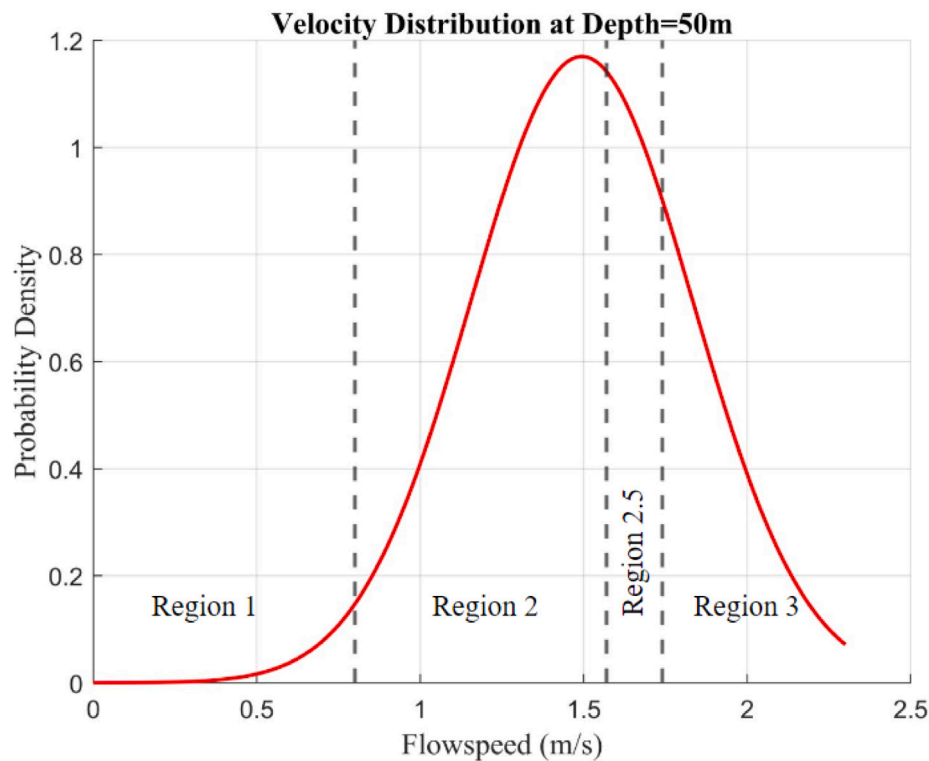


Fig. 18. Flow speed distribution measured along the 27°N transect line with a bottom-mounted ADCP.

For an OCT depth of 50 m, the calculated closed-loop AEP for this turbine, based on the defined power and distribution curves, is approximately 171.25 MWh per year.

5. Conclusions

This research explores the process of design and testing of a grid-connected hardware-in-the-loop ocean current turbine using an OPAL-RT real-time HIL simulator. An OCT model was executed in real-time while operating in realistic hurricane conditions using data gathered from ADCP readings. This model was then connected to a microgrid, where the grid connection performance was evaluated and found to have stable power and frequency while operating in Region 3 with a blade pitch controller implemented to reduce the power generated as the turbine experiences increasing flow speeds.

The model OCT was then replaced by a 2-HP dynamometer in order to complete the HIL test bench. The HIL feedback was first validated to ensure that the simulation was receiving accurate readings. A grid-connected case study was performed by scaling the dynamometer torque, and a variable speed controller operating in Region 2 was implemented. The results showed that the controller maintains the desired power coefficient under increasing flow speed changes, while the grid power and frequency remain stable. Additionally, a controller was designed for Region 2.5 to ensure a smooth transition between operational regions. Testing on the HIL testbed demonstrated its effectiveness, showing accurate and reliable performance by both the controller and the testbed. The flexibility of HIL not only enhances the testbed's relevance as technology evolves but also opens the door to future works of conducting advanced tests, such as cybersecurity assessments, where real-time interactions between hardware and control systems are critical. The ability to run simulations in real-time with actual hardware ensures that the testbed can serve as a foundational tool for both current research and future developments in the field of ocean energy.

The development of a HIL test bench for an ocean current turbine coupled with a simulated grid connection represents an advancement in marine renewable energy research, with a particular focus on marine current energy applications including tidal and ocean currents. This approach offers a controlled and efficient means of evaluating the performance and reliability of ocean current turbines in a simulated environment, minimizing risks and costs associated with real-world testing. Considering coastal communities, small island developing states, and the blue economy, this HIL test bench not only accelerates technological advancements but also contributes to the sustainable energy solutions necessary for addressing the diverse energy needs of these regions.

CRediT authorship contribution statement

Sasha Fung: Writing – original draft, Investigation, Formal analysis, Data curation. **Yufei Tang:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Carter Nichols:** Writing – original draft, Visualization, Investigation, Data curation. **James VanZwieten:** Writing – review & editing, Supervision, Conceptualization. **Hassan Mokari:** Methodology. **Gabriel Alsenas:** Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Yufei Tang reports financial support was provided by U.S. National Science Foundation. Yufei Tang reports financial support was provided by U.S. Department of Energy (DOE). If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] M. Melikoglu, Current status and future of ocean energy sources: A global review, *Ocean Eng.* 148 (2018) 563–573.
- [2] X. Yang, K.A. Haas, H.M. Fritz, Theoretical assessment of ocean current energy potential for the gulf stream system, *Mar. Technol. Soc. J.* 47 (4) (2013) 101–112.
- [3] N. Rahman, S. Badshah, A. Rafai, M. Badshah, Literature review of ocean current turbine, *Int. J. Sci. Eng. Res.* 5 (11) (2014) 11.
- [4] B. Davis, J. Farrell, D. Swan, K. Jeffers, Generation Of Electrical Power From The Florida Current Of The Gulf Stream, in: OTC Offshore Technology Conference, All Days, OTC-5120-MS, 1985, <http://dx.doi.org/10.4043/5120-MS>, arXiv:<https://onepetro.org/OTCONF/proceedings-pdf/86OTC/All-86OTC/OTC-5120-MS/2031151/otc-5120-ms.pdf>.
- [5] X. Yang, K.A. Haas, H.M. Fritz, Evaluating the potential for energy extraction from turbines in the gulf stream system, *Renew. Energy* (ISSN: 0960-1481) 72 (2014) 12–21, <http://dx.doi.org/10.1016/j.renene.2014.06.039>, URL <https://www.sciencedirect.com/science/article/pii/S0960148114003796>.
- [6] K. Haas, Assessment of Energy Production Potential from Ocean Currents Along the United States Coastline, Georgia Tech Research Corporation, 2013.
- [7] J. Stewart, H. B. (Eds.), Proceedings of the MacArthur Workshop on the Feasibility of Extracting Useable Energy from the Florida Current, National Technical Information Service, U.S. Department of Commerce, Palm Beach Shores, Florida, 1974, Reproduced by the National Technical Information Service.
- [8] D.T. Mouton Jr., Submarine Turbine Power Plant, Mouton US Patent, 1980, p. 4,219,303, URL <https://patents.google.com/patent/US4219303A/en>.
- [9] F. Akwensivie, In the wake of a marine current turbine, The University of Strathclyde, 2004.
- [10] P. Pyakurel, J.H. VanZwieten, M. Dhanak, N.I. Xiros, Numerical modeling of turbulence and its effect on ocean current turbines, *Int. J. Mar. Energy* (ISSN: 2214-1669) 17 (2017) 84–97, <http://dx.doi.org/10.1016/j.ijome.2017.01.001>.
- [11] J.H. VanZwieten, N. Vanrietvelde, B.L. Hacker, Numerical simulation of an experimental ocean current turbine, *IEEE J. Ocean. Eng.* 38 (1) (2013) 131–143, <http://dx.doi.org/10.1109/JOE.2012.2218891>.
- [12] P. Razi, P. Ramaprabhu, P. Tarey, M. Muglia, C. Vermillion, A low-order wake interaction modeling framework for the performance of ocean current turbines under turbulent conditions, *Renew. Energy* (ISSN: 0960-1481) (2022) <http://dx.doi.org/10.1016/j.renene.2022.10.001>, URL <https://www.sciencedirect.com/science/article/pii/S096014812201494X>.
- [13] J. Khan, Equivalent time-domain water-to-wire model of a generic tidal current turbine, in: 2021 IEEE Power & Energy Society General Meeting (PESGM), 2021, <http://dx.doi.org/10.1109/PESGM46819.2021.9637955>.
- [14] Y. Tang, J. VanZwieten, B. Dunlap, D. Wilson, C. Sultan, N. Xiros, In-stream hydrokinetic turbine fault detection and fault tolerant control-a benchmark model, in: 2019 American Control Conference (ACC), IEEE, 2019, pp. 4442–4447.
- [15] S. Seo, J. Kim, E. Muljadi, S. Meor-Danial, M. Worthington, R. Wills, Reactive power control to mitigate flicker in a tidal and river hydrokinetic power generation, in: 2019 IEEE Power & Energy Society General Meeting (PESGM), 2019, <http://dx.doi.org/10.1109/PESGM40551.2019.8973527>.
- [16] K. Tsuji, M. Shiono, Effect of stator D-axis current for tidal current power-generation system using constant turbine-output control scheme, in: 2020 IEEE Power & Energy Society General Meeting (PESGM), 2020, <http://dx.doi.org/10.1109/PESGM41954.2020.9281556>.
- [17] J.H. VanZwieten, P. Pyakurel, T. Ngo, C. Sultan, N.I. Xiros, An assessment of using variable blade pitch for moored ocean current turbine flight control, *Int. J. Mar. Energy* (ISSN: 2214-1669) 13 (2016) 16–26, <http://dx.doi.org/10.1016/j.ijome.2016.01.002>, URL <https://www.sciencedirect.com/science/article/pii/S2214166916000035>.
- [18] V. Chalisahaz, S. Bhattacharya, S. Hanif, D. Bhatnagar, M.J. E. Alam, B. Robertson, D. Prezioso, R. O'Neil, Data-driven reliability assessment for marine renewable energy enabled island power systems, in: 2021 IEEE Power & Energy Society General Meeting (PESGM), 2021, <http://dx.doi.org/10.1109/PESGM46819.2021.9637995>.
- [19] B. Freeman, Y. Tang, Y. Huang, J. VanZwieten, Rotor blade imbalance fault detection for variable-speed marine current turbines via generator power signal analysis, *Ocean Eng.* 223 (2021) 108666.
- [20] B. Freeman, Y. Tang, Y. Huang, J. VanZwieten, Physics-informed turbulence intensity infusion: A new hybrid approach for marine current turbine rotor blade fault detection, *Ocean Eng.* 254 (2022) 111299.
- [21] M. Neshati, P. Feja, A. Zuga, H. Roettgers, A. Mendonca, J. Wenske, Hardware-in-the-loop testing of wind turbine nacelles for electrical certification on a dynamometer test rig, *J. Phys. Conf. Ser.* 1618 (3) (2020) 032042, <http://dx.doi.org/10.1088/1742-6596/1618/3/032042>.
- [22] A. Fontanella, A. Facchinetti, E. Daka, M. Belloli, Modeling the coupled aero-hydro-servo-dynamic response of 15 MW floating wind turbines with wind tunnel hardware in the loop, *Renew. Energy* 219 (2023) 119442.
- [23] L.K. Gan, E.J.P.E. Subiabre, A realistic laboratory development of an isolated wind-battery system, *Renew. Energy* 136 (2019) 645–656.
- [24] R.F. Schkoda, C. Fox, R. Hadidi, V. Gevorgian, R. Wallen, S. Lambert, Hardware-in-the-loop testing of utility-scale wind turbine generators, 2016.
- [25] A. Poureh, A. Nobakhti, Robust control design for an industrial wind turbine with HIL simulations, *ISA Trans.* 103 (2020) <http://dx.doi.org/10.1016/j.isatra.2020.05.004>.
- [26] G. Caraiman, C. Nichita, V. Mi nzu, B. Dakyo, Marine current turbine emulator design based on hardware in the loop simulator structure, in: Proceedings of 14th International Power Electronics and Motion Control Conference EPE-PEMC 2010, 2010, <http://dx.doi.org/10.1109/EPEPEMC.2010.5606593>.
- [27] F. Huerta, J.K. Gruber, M. Prodanovic, P. Matatagui, A power-HIL microgrid testbed: Smart energy integration lab (SEIL), in: 2014 IEEE Energy Conversion Congress and Exposition (ECCE), 2014, pp. 3998–4003, <http://dx.doi.org/10.1109/ECCE.2014.6953945>.
- [28] M. Ouhrouche, Transient analysis of a grid connected wind driven induction generator using a real-time simulation platform, *Renew. Energy* (ISSN: 0960-1481) 34 (3) (2009) 801–806, <http://dx.doi.org/10.1016/j.renene.2008.04.028>, URL <https://www.sciencedirect.com/science/article/pii/S096014810800164X>.
- [29] V. Ruuskanen, J. Koponen, T. Sillanpää, K. Huoman, A. Kosonen, M. Niemelä, J. Ahola, Design and implementation of a power-hardware-in-loop simulator for water electrolysis emulation, *Renew. Energy* (ISSN: 0960-1481) 119 (2018) 106–115, <http://dx.doi.org/10.1016/j.renene.2017.11.088>, URL <https://www.sciencedirect.com/science/article/pii/S0960148117311898>.
- [30] A. Hasankhani, J.H. VanZwieten, Y.-S. Tang, B. Dunlap, A.D. Luera, C. Sultan, N.I. Xiros, Modeling and numerical simulation of a buoyancy controlled ocean current turbine, *Int. Mar. Energy J.* (2021).
- [31] W. Laing, J.H. VanSwieten, et al., Development of a driving electric dynamometer rotor emulator for MHK in-stream turbines, 2014.
- [32] M.L. Buhl, WT _ Perf user's guide, NWTC Des. Codes (2004).
- [33] MathWorks, Per-Unit and International Systems of Units, URL <https://www.mathworks.com/help/sps/powersys/ref/per-unit-and-international-systems-of-units.html>.
- [34] R. Reid, R. Saulnier, Gagnon, Wind-turbine asynchronous generator in isolated network, The MathWorks, Inc 2022b (2022).
- [35] S. Fung, Y. Tang, J. VanZwieten, G. Alsenas, Modeling and Real-Time Simulation of Ocean Current Turbines for Grid Integration, 07, 2023, pp. 1–5, <http://dx.doi.org/10.1109/PESGM52003.2023.10252303>.
- [36] L. Lee, J. VanZwieten, Adaptive individual blade pitch control of an ocean current turbine for load reduction, *FAU Undergrad. Res. J.* 6 (2017) 51–51.
- [37] Wind turbine, the mathworks, inc 2023b, 2023, URL <https://www.mathworks.com/help/sps/ug/wind-turbine.html>.
- [38] Y.-b. Liang, L.-x. Zhang, E.-x. Li, F.-y. Zhang, Blade pitch control of straight-bladed vertical axis wind turbine, *J. Cent South Univ.* 23 (5) (2016) 1106–1114.
- [39] Q. Hawari, T. Kim, C. Ward, J. Fleming, A robust gain scheduling method for a PI collective pitch controller of multi-MW onshore wind turbines, *Renew. Energy* (ISSN: 0960-1481) 192 (2022) 443–455, <http://dx.doi.org/10.1016/j.renene.2022.04.117>, URL <https://www.sciencedirect.com/science/article/pii/S0960148122005857>.
- [40] P.W. Carlin, A.S. Laxson, E.B. Muljadi, The history and state of the art of variable-speed wind turbine technology, *Wind Energy* 6 (2) (2003) 129–159, <http://dx.doi.org/10.1002/we.77>, URL <https://onlinelibrary.wiley.com/doi/abs/10.1002/we.77>, arXiv:<https://onlinelibrary.wiley.com/doi/pdf/10.1002/we.77>.
- [41] B. Boukhezzer, L. Lupu, H. Siguerdijane, M. Hand, Multivariable control strategy for variable speed, variable pitch wind turbines, *Renew. Energy* (ISSN: 0960-1481) 32 (8) (2007) 1273–1287, <http://dx.doi.org/10.1016/j.renene.2006.06.010>, URL <https://www.sciencedirect.com/science/article/pii/S09601481060001261>.
- [42] B. Yang, T. Yu, H. Shu, Y. Zhang, J. Chen, Y. Sang, L. Jiang, Passivity-based sliding-mode control design for optimal power extraction of a PMSG based variable speed wind turbine, *Renew. Energy* (ISSN: 0960-1481) 119 (2018) 577–589, <http://dx.doi.org/10.1016/j.renene.2017.12.047>, URL <https://www.sciencedirect.com/science/article/pii/S096014811731251X>.
- [43] F.A.A. Nader Majed Moustafa, A.F. Nori, PID control system for a variable speed horizontal axis wind mill, *Int. J. Mech. Eng. Technol.* (2018) 1080–1087.
- [44] E.A. Bossanyi, The design of closed loop controllers for wind turbines, *Wind Energy* 3 (3) (2000) 149–163, <http://dx.doi.org/10.1002/we.34>, URL <https://onlinelibrary.wiley.com/doi/abs/10.1002/we.34>, arXiv:<https://onlinelibrary.wiley.com/doi/pdf/10.1002/we.34>.
- [45] T. Jeon, D. Kim, I. Paek, Improvements to and experimental validation of PI controllers using a reference bias control algorithm for wind turbines, *Energies* (ISSN: 1996-1073) 15 (21) (2022) <http://dx.doi.org/10.3390/en15218298>, URL <https://www.mdpi.com/1996-1073/15/21/8298>.
- [46] W. Laing, J.H. VanZwieten, Development of a driving electric dynamometer rotor emulator for MHK in-stream turbines, 2014.
- [47] K.E. Johnson, Adaptive torque control of variable speed wind turbines, ProQuest Diss. Theses (2004) 142.
- [48] J.P. Cangialosi, A.S. Latto, R. Berg, Tropical cyclone report: Hurricane Irma (AL112017) 30 august–12 september 2017, in: National Oceanic and Atmospheric Administration (NOAA), 2018, National Hurricane Center. [Online]. Available <https://www.nhc.noaa.gov/data/tcr/index.php?season=2017&basin=atl>.

- [49] National Centers for Environmental Information, Marine beaufort scale, 2021, Accessed: 2024-05-10. URL https://www.ncei.noaa.gov/sites/default/files/2021-09/Marine_Beaufort_Scale.pdf.
- [50] E.B. Muhando, T. Senjyu, E. Omine, H. Kinjo, T. Funabashi, Model development for nonlinear dynamic energy conversion system: an advanced intelligent control paradigm for optimality and reliability, *IEEJ Trans. Electr. Electron. Eng.* 3 (5) (2008) 482–491, <http://dx.doi.org/10.1002/tee.20303>, URL <https://onlinelibrary.wiley.com/doi/pdf/10.1002/tee.20303>, arXiv:<https://onlinelibrary.wiley.com/doi/pdf/10.1002/tee.20303>.
- [51] E.B. Ondes, C. Sultan, A. Hasankhani, J.H. VanZwieten, N.I. Xiros, ν -Gap metric based multi-model predictive control of an ocean current turbine system with blade pitch failures, *Ocean Eng.* (ISSN: 0029-8018) 278 (2023) 114201, <http://dx.doi.org/10.1016/j.oceaneng.2023.114201>, URL <https://www.sciencedirect.com/science/article/pii/S0029801823005851>.
- [52] M.A. Abdelbaky, X. Liu, D. Jiang, Design and implementation of partial offline fuzzy model-predictive pitch controller for large-scale wind-turbines, *Renew. Energy* (ISSN: 0960-1481) 145 (2020) 981–996, <http://dx.doi.org/10.1016/j.renene.2019.05.074>, URL <https://www.sciencedirect.com/science/article/pii/S0960148119307475>.
- [53] M.C.P.M. Machado, J.H. VanZwieten, A.P.O. Callou, I. Pinos, A measurement based analyses of the hydrokinetic energy in the gulf stream, in: *International Ocean and Polar Engineering Conference, 2015, All Days ISOPE-I-15-491*. arXiv:<https://onepetro.org/ISOPEIOPEC/proceedings-pdf/ISOPE15/All-ISOPE15/ISOPE-I-15-491/1334920/isope-i-15-491.pdf>.